

Impact Evaluator Design - Juan Benet



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The benefits of markets in altruistic impact, and what's needed to achieve them Funding the Commons Conference Learn more at <https://fundingthecommons.io/> 0:00 Introduction 1:14 Foundational ideas 2:50 Benefits of markets 9:25 Basic requirements for impact markets 9:31 Bookkeeping 10:20 Social buy-in 12:25 Infrastructure 14:24 Critical path? 15:20 Key downside risks 18:46 Goodhart's law 19:56 Unknown unknowns 20:47 Verici 0.1 21:02 Norms for markets 24:14 Verifiability 25:19 Rights 28:48 Consumption 30:34 Incremental

Introduction

0:02

and i'd like to welcome our next speaker owen cotton-barrett is a research

0:07

associate in the future of humanity institute at the university of oxford focusing on decisions in the face of

0:14

great uncertainty he's an effective altruist director of the research at the center for effective

0:19

altruism and holder of a d field in mathematics owen will be speaking to us about the

0:25

about impact certificates and impact markets the benefit of markets and altruistic impact and what's needed to

0:32

achieve them welcome owen uh hi i'm really happy to be here so i

0:38

i'm going to talk about uh impact certificates and impact markets i think that many of the original uh ideas

0:44

here originally traced back to uh paul christiano and dennis drescher has recently been helping me uh

0:51

work out some things so this is a kind of relatively uh theory talk about how we could be

0:57

reconceptualizing uh impact and how um the doing of good things uh we relate

1:04

to that and what types of things even can be tradable um but i think there might be a lot of value on the table by

1:11

getting a good system set up here so what what are we even talking about uh roughly i'm thinking of impact is

Foundational ideas

1:20

uh managing to change in the supply of a public or maybe private good um

1:27

there's a couple of modifiers here uh if you ask change what does it that's supposed to be change relative to

1:34

the idea is that there's some kind of counterfactual world history where you didn't take a particular action um

1:40

mostly i think that we can not bother too much with this and we just have this intuitive notion of like oh you know

1:46

what what do we mean by by impact and uh that'll flow through okay but i think

1:51

it's interesting to uh be able to bottom out what we mean

1:59

if we need to and then i also think that i'm regarding it as the responsibility for that change

2:04

and that the key notion is that um you like do something good and then the the impact

2:10

flows back to the person responsible um so that's the basic setup

2:16

uh and then maybe the kind of like an interesting question is well if this is a thing that belongs to people often for things that

2:22

belong to people we assign people property rights over them um and they can say oh well you know i want to give

2:28

this away i want to assign this to to somebody else and then you can start building markets on top of

2:34

that um and we can ask the question for impact as we've defined it here should we be able to uh

2:41

give assignation rights and have people say actually this impact which was coming to me i am passing on uh to another party

2:48

um and if we do then that could be the first step towards allowing markets in

Benefits of markets

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impact um uh like impact is this slightly rarefied uh

3:01

abstract notion and i think that there are several possible benefits of having

3:07

markets and they're kind of the usual benefits but transferred over to a new context i

3:13

think that several of these benefits we can get through uh mechanisms which don't make any

3:19

reference to uh uh like rarefied impact like i think

3:25

that i've put here as the first benefit ex-post evaluation of a

3:30

charitable work or other other work that's aiming uh to be good um uh and prizes are a normal way of doing that

3:38

but i i think that uh the reconceptualization can help uh to spot places where we can

3:45

uh port over other market mechanisms and and there can be something useful about it

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so yeah the normal story for why x post evaluation is better is that

3:56

uh you you have more information like we you just get to see more oh did that turn out well um uh versus if you're uh

4:04

restricted to funding things uh beforehand then you have to go on the basis of do people have a plausible

4:11

story and private information is um

4:16

like bites harder in that case uh i think that

4:22

yeah so uh this can allow for better integration of diffused information where um

4:29

somewhere before kind of the final price for some set uh the place allow whichever individuals

4:37

have pertinent information as to uh the eventual value of something to uh

4:43

get that information into the market uh by buying or selling and

4:49

uh the markets can be a good way of integrating that information

4:56

and then i think another potentially big deal is that it can incentivize excellence if you have a

5:03

system where uh markets can give big rewards to people who do

5:09

things that are really good then it incentivizes people to look for hey how could we do something that's great um

5:15

and i think that uh just humans are often incentivized by a

5:23

things like uh getting recognition but also getting rich and uh there's a lot of energy that goes

5:29

into um oh how can we make a for-profit company that's that's gonna uh uh make

5:35

us rich and i think that if we could get that um same innovation and energy going

5:40

into uh how can we do things where a for-profit model doesn't work but we

5:45

still want to be really ambitious and aim for uh big changes in the world then uh uh this is something where markets

5:52

and impact can be pretty helpful um so the basic i

5:57

like conceit uh which i'm gonna illustrate through to a metaphor here and so the

6:03

idea is like rather than just accepting the impact from a good action that you've done and and you in this case

6:08

could be a person or it could be an organization uh or maybe it can be some other kind of uh

6:14

entity but you you uh like intercept with a bottle um and you say no i'm not

6:20

taking the impact for this instead i'm going to put it into storage and now it's a transferable unit which i can go

6:26

and sell um and yeah so you can think of this like an nft for for

6:32

a particular good deed that has been done um and

6:37

the metaphor with the bottles uh the key step here is that uh owners can later

6:44

open the bottle um a kind of und uh meaning that now whoever

6:50

uh gets to uh drink it or whatever gets the

6:56

impact that was originally bottled and so um if you have tokens then at this point

7:01

they would become untradeable they would record uh who

7:07

uh was attributed the uh

7:12

associated impact um and maybe if people often think of nfts as

7:19

like a market in fine art this might be sort of more like a market in fine wine

7:24

uh ultimately uh the value of things is propped up by people wanting to uh

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consume them at some future point um so that's the basic idea and then

7:37

there's lots of reasons why this is pretty hard to get set up uh one is that impact attribution is often

7:44

implicit and it's complicated you have a thing which a bunch of people have collaborated on and it's just not clear

7:51

who gets which amount of credit um and and it's a mess to try and sort out and that means

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that it's a mess for anybody to try and uh sell the associated impact

8:02

because the buyers don't really know what they're getting um i also think that right now we don't

8:09

have good ways of recording anything there um

8:16

and like the recording thing is we can deal

8:21

with and then finally you know nobody cares we don't have people who are saying oh yeah i really want to buy into

8:28

these impact markets there isn't the demand for it um

8:33

so one other concept is i want to draw a distinction between impact certificates which are the things i was representing by bottles tokens uh denoting the rights

8:43

to attribute the impact associated with uh some particular uh uh specified actions

8:49

uh and impact shares which entitle uh the owners to share a future stream of

8:54

impact uh uh certificates from whatever their own entities uh i mean impact shares are

9:00

just kind of shares uh but in in the impact context i just wanted to draw this distinction because i think that

9:05

sometimes when i've heard people uh talking about ideas here uh

9:12

the two can be blurred and i think that conceptually they're distinct conceptually they're both useful and i

9:18

expect that if we ever do get to the point where we're building mature ecosystems around these we will be using both

Basic requirements for impact markets

9:25

so what would be needed to set up impact markets i think there's uh maybe three central

Bookkeeping

9:32

things there's uh bookkeeping we want some way of uh

9:38

recording okay who owns the impact associated with particular uh actions um

9:44

uh such that people can uh trust uh however that bookkeeping process is done

9:50

um i i think that uh kind of connected to the bookkeeping is you want some basic way to validate

9:57

that the the records uh of the things that are being written down line up with

10:02

real world impact that they are uh supposed to represent um uh you don't want somebody just

10:08

claiming hey i did this amazing thing um uh when uh and and getting paid for it when uh

10:15

uh it's kind of if essentially just fraudulent um i think that the second is needing

Social buy-in

10:23

social buying you need buy-in from lots of different parties you want buy-in from whichever bodies are potentially

10:29

doing the uh good deeds which would be bottled and issued as certificates um

10:36

you need buy-in from uh marketplaces to facilitate

10:43

uh the uh uh purchase of these um uh but i

10:48

think and and maybe buying from

10:53

if you want to get benefits of uh speculators and markets who maybe want the buy-ins uh from

10:59

uh speculators but i think that the key thing is from the ultimate buyers uh those people who are uh have an interest

11:06

in eventually consuming um uh the bottled impact and so

11:11

here i might think of this as

11:17

substituting in for when people currently give money to uh charities or

11:22

other uh good causes um and saying well hey maybe they would

11:29

prefer to buy uh impact you know if you're giving money to a charity

11:35

i think that the social contract is such that implicitly you are buying some impact from the charity you're saying

11:40

well i uh want to give this money because i trust that it will be used for good things um

11:48

and this is saying well there's a different setup where you could uh still

11:53

buy the good deeds but you get to find out exactly what you've bought rather than doing it uh all on trust that

11:59

something uh good will happen um also if these things get implemented

12:05

legally i think that the point of consumption of these certificate

12:11

should be tax deductible uh there's a bunch of questions of how you can actually set

12:17

that up but that's the stage which is uh very tightly analogous to making

12:22

donations um and then there's a bunch of infrastructure uh that one would want to

Infrastructure

12:28

have in a mature ecosystem uh uh helping to support this

12:34

um i think maybe the

12:41

most key elements here the ones three certificate auditors you want um

12:46

some bodies which do the work of investigating

12:52

um certificates to check out and say hey yeah this is legit the thing which it is

12:58

claiming happened really did happen um because if

13:03

that is left to each individual buyer to do uh then that presents a massive amount

13:09

of overhead on the market and uh is likely to uh make

13:15

uh uh just yeah purchases unwilling to enter

13:21

and maybe you also early on want to have uh

13:26

certificate issuers which help to facilitate that process it's also possible that you could set it up in

13:32

such a way that people self-issue you might want to combine uh three and four here you might

13:37

want to have some uh relatively trusted bodies which arise as uh hey we will

13:44

issue certificates and uh uh we will develop a reputation which

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people uh get trust in that we do a good job of auditing the things that we issue certificates for

13:57

and therefore any certificate issued by us can be trusted um but that could also uh just live as a layer on
uh

14:05

on top of this i'm imagining that a mature market would have competition between uh multiple different
auditors

14:11

and uh trying to audit things efficiently without spending too much resources but also uh to a high enough

14:19

standard that it establishes a high degree of trust and then there's a question of okay well

Critical path?

14:25

if we would eventually need all of these things what is the critical path uh towards setting up markets here

14:31

um and of the three bookkeeping social buying and infrastructure i really think that social buy-in is

14:39

uh uh the key aspect uh we want people using this and then that will

14:45

the more that there's demand and people using it the more that will drive everything else uh uh

14:51

being built to meet that demand but i think that

14:56

um uh bookkeeping and uh some aspects of infrastructure are helpful for

15:02

facilitating uh uh that social buy-in if there's a clean implementation if people feel like oh yeah okay uh it's it's very

15:10

obvious what's happening and i feel trust in the technical way that's been set this has been set up then that can facilitate

15:16

the social buy-in but i don't think that it's the only part of social buying which matters um

Key downside risks

15:24

i outlined earlier why i am excited about impact markets what i think that the the big upsides here are

15:31

i also think that there are uh a number of

15:37

potential downsides and i'm just going to highlight a handful

15:43

which seem to me like the biggest issues um one is

15:50

that at least on the a naive implementation of this uh impact markets could incentivize likely

15:57

harmful charities and the reason is that there's an asymmetry where if a charity uh

16:03

does a bunch of work i mean sorry i'm i'm saying charity but it could just be uh actions uh which

16:11

uh somebody is doing where they're hoping to make

16:16

uh money by selling the impact um yeah if somebody does something that's

16:22

very risky and that there's uh some chance of getting uh extremely bad

16:28

outcomes and some chance of getting extremely good outcomes um

16:33

the value of the associated certificates is generally capped below by zero

16:39

um it's due to know zero is what you get if nobody in the market is willing to pay for this um

16:45

and this could mean that uh if there's a high tail upside that

16:50

people are incentivized to take those actions uh even if the

16:55

expected value of the thing is strongly negative

17:00

now this isn't a problem which is unique to impact markets uh in general it's a

17:08

question of negative externalities and there's a question of how society should uh coordinate to

17:14

uh try to tax negative externalities or

17:20

otherwise uh incentivize people to uh not create them um

17:25

but i do think that it can bite the issue can bite particularly cleanly here and so it's worth us some extra

17:32

attention yeah it's worth pointing out that um you might think well uh markets

17:38

generally let people send a price signal uh that they they think something is

17:44

overvalued by shorting a stock uh but because the eventual resolution here

17:50

can't go below zero that doesn't provide a direct way for um uh markets to say

17:56

that they think that something is uh actually strongly net negative so this could be this could be worked on

18:03

um and maybe there are some uh interesting

18:10

innovative solutions which could work here this could be regarded as

18:15

paul christiani pointed this out to me part of the general problem uh that impact markets

18:22

uh as we're currently conceiving them don't solve the problem of moral trade

18:28

where you uh meaning that you fail to reach pareto

18:34

superior outcomes uh which everybody would um prefer

18:40

because then they're nobody's kind of first preferred option uh by trade

Goodhart's law

18:46

um i think that there's also issues about changing the incentives uh otherwise on

18:54

um people pursuing good work and it might push people towards things which are

19:00

easier to uh turn into uh impact that can be sold

19:06

um or easier to uh or to put more work into

19:13

marketing and explaining why this thing that you did in the past was really good of course this is to some extent a duel

19:18

of issues which already come up for charities where they can be incentivized

19:24

to pursue work that they think uh will uh

19:29

sell well and they'll be able to persuade people to give them money to do rather than for having done

19:35

um and uh to put a lot of money into fundraising for um uh

19:41

uh uh their future work so there are traps there it's not

19:46

um i don't have a particular reason to think that they are larger than the current issues that uh

19:53

could be worth for the investigation and in general there's a kind of like hesitance of well

Unknown unknowns

20:00

this could be a large shake up in how

20:05

uh charitable work is funded and uh some people have been thinking about

20:10

that some but uh we definitely haven't worked through all of the implications of that and maybe there are unknown effects of how it affect uh of

20:18

the impacts on on the ecosystem uh of moving to this system so i'm

20:23

this one makes me particularly interested in uh experiments which have some ability

20:30

to say okay no let's let's close up shop uh this uh seems like it's not going uh in

20:37

a in a productive direction um uh but there are complications of how that

20:43

interacts with incentivizing trust from people okay and so then in the last part of the

Verici 0.1

20:49

talk i want to talk about what uh we're calling virici 0.1 um

20:54

the numbering system and name is to to suggest that this really is uh kind of

21:00

early stage thinking and the idea here is that if we want functional markets and we

Norms for markets

21:06

want people uh having a lot of trust in those markets i think that that

21:11

depends on people having a good understanding of how the markets work

21:18

um and if i go and i try to buy a stock on an

21:25

exchange i don't need to have gone and looked into all of the legal contracts that are

21:30

supporting the particular um uh type of stock that i

21:36

am buying and to check and i i can just have like some generalized trust in the exchange

21:42

and uh that the thing is regulated in some way which

21:50

uh uh stops them selling uh 50 times as much

21:55

stock as actually exists etc um

22:01

and i think that there are other ways if we just set

22:06

set up common expectations of how things are going to work that could be pretty helpful and so

22:12

this is my current best guess uh for the social norms uh that we should try to

22:20

set up as the rules of markets uh impact markets if we were doing experiments

22:25

today to establish impact markets and

22:32

uh you know i'm sure there'll be mistakes in here but i really think it is worth trying if we're making attempts to start with

22:39

our best guesses and so there are a few different properties the first is verifiability that impacts has

22:44

occurred i think that it is helpful if there is gatekeeping around the market

22:52

which makes it clear to market entrance that uh uh things which are on sale there

22:58

have passed some level of auditing uh that they really happened um

23:04

i think that it's also really important to have clear ownership of uh

23:10

rights and not to try and sell impact in cases where there's too much ambiguity uh over ownership rights

23:17

i think that we want some kind of consumption mechanism uh this equivalent

23:23

of uh unbottling uh the impact treating it like fine wine rather than fine art

23:29

um and i think that we want to set things up in such a way that they

23:35

are incremental and that this is acceptable to

23:41

existing uh interests uh in in the markets

23:46

um and so i'm not going to say a bit more about each of these four

23:52

and yeah they they don't yet address most of the risks i mentioned before um these are more

23:59

uh rules where i think creating common knowledge around this uh would help give

24:04

the best shot of creating healthy

24:10

markets which a lot of people are actually happy buying into um

Verifiability

24:16

yeah i mean the basic issue with verifiability is that it's hard to facilitate trust and buyers if people

24:22

can't verify whether the impact has happened um and

24:28

uh as a matter of ui and making the process

24:34

relatively seamless for buyers i think it's helpful if there is some relatively trusted zone where they can operate in

24:40

and they can say well now i'm just going to look at what the things are claiming rather than thinking any time i need to make a

24:46

purchase here i i need to uh go and do a massive amount of uh

24:52

due diligence and there are various different ways that this could be set up it could be um

24:58

uh that one uh has a marketplace and one accredits

25:05

various different auditors and then uh buyers coming in can do filters on i

25:11

only want to see things which have been approved by uh this kind of auditor

25:17

um yeah so the issue of rights i think is

Rights

25:24

interesting i think that there are lots of cases where uh say i work on a project with someone

25:31

um and then we say and then then we're like oh well that did well and now i want to go and sell the impact uh for my

25:39

work on this this can create adversarial dynamics uh and incentives between the

25:44

people who were the collaborators because now they're each incentivized to produce a public narrative uh for why

25:51

their contributions were particularly important and other people's contributions uh were relatively

25:56

unimportant and i think that that could be um pretty unhealthy and destructive

26:02

and so i think that we uh want to avoid having that and

26:08

often i think that the way to do this will be via explicit agreements among collaborators

26:15

about uh how impact will be shared um uh similar to the advice for

26:22

uh startup founders to uh agree a share of equity very early on

26:30

uh in in the process so and society has to grapple with it with

26:35

this uh uh in in various cases and there were um a bunch of situations where uh ownership

26:42

already ends up uh pretty clear and like society has allocated a standard answer

26:48

to this if somebody owns a company uh a 100 then presumably they own uh

26:54

all of the uh uh impact certificates that that company is generating um uh similarly

26:59

for copyright and if there is uh impact associated uh uh

27:05

uh coming from the uh the copyrighted material um

27:12

by default i think that should go to the copyright owner um i think that

27:19

one can specify uh clear explicit divisions uh and in general if you're doing something by yourself uh it's uh

27:26

pretty simple relatively unclear cases include uh donating to a charity and so

27:32

uh a challenge for getting impact markets off the ground is that i think

27:38

that right now charities often couldn't uh in good faith issue

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certificates of impact because there is significant ambiguity as to how much the

27:51

rights for the good deeds that they have done lies with the charity rather than the donors to the charity if they have

27:58

some kind of um uh explicit expectation setting with the donors uh uh then i

28:03

think we'll clear that up um i think

28:08

you get a similar thing with um people who are employed by a charity

28:16

where i think that there's an implicit social contract that uh they

28:23

are doing that in part because they regard their work as contributing to the good

28:30

that the charity is doing uh but i think it's ambiguous as to how big their contribution is

28:35

and just lots of other activities where there's a collaboration uh but there's no been no explicit discussion about

28:41

divisional credit um

28:47

then consumption uh i was pointing to earlier i think that this has a couple of different benefits

Consumption

28:52

one is that it helps to provide social proof um that there are ultimate buyers uh interested uh and these buyers are

29:00

about removing tokens from circulation and so i think that this should uh

29:05

support trust in underlying value of uh

29:11

tokens because it builds up a track record um and less than fears of bubbles i also think that it helps with uh the

29:18

the final property of being incremental because it provides continuity with the current practice um

29:24

in the current practice uh implicitly when somebody is doing

29:30

something good they're just consuming uh the impact immediately and this is saying hey there's another

29:36

operation that they could take uh where they say no we're going to hold off on that um and bottle it and put it into a

29:43

certificate and then maybe that can be sold but if they don't do that uh uh we're just in the status quo one

29:50

um and i think that that continuity is helpful because if you imagined uh

29:55

something without a property like this then there could be uh a bunch of people saying hey does this

30:02

impact markets thing mean that now lots of charities need to be paid again for

30:07

the work they did didn't we already give them money for that uh ahead of time and i think that that

30:14

could create bad feeling and could uh make

30:19

parties uh who like the parties who would be giving money here uh uh somewhat

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disinclined to engage with the process for fear of being exploited

30:33

um yeah so this is the the basic principle behind

Incremental

30:38

incremental i think that at every stage we want to uh

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set things up so there's just incentives for uh people to uh come in and start using

30:50

this uh and that nobody is directly losing out as a

30:56

result yeah in order to avoid creating uh kind of entrenched opposition uh uh

31:02

to the system i think that uh this can be totally

31:07

i i think this is totally achievable um and i think that uh it's normally the good pathway to to go for if uh you

31:16

want a new economic technology to to to take over um okay so uh i'll wrap up that uh just

31:24

to as as and then we can take a few questions and the

31:29

final comment is this yeah i kind of hopeful that impact markets

31:34

uh could be a big thing and could transform um

31:40

how a lot of analysis is done and a lot of incentives in the ecosystem of people trying to do good

31:46

things and i'm really interested to to see how they develop and i

31:52

i think that this kind of conference is great because the people at conferences like there so exactly

31:58

the right people to be going and building experiments well thanks owen for a comprehensive

32:05

look at markets and altruistic impact that acknowledges the risks as well as the upsides and we have some time for

32:12

questions and i know there are a few in the chat if you don't mind i'll service them here the first question

32:18

is from actually another one of our uh forthcoming presenters andrew kritsch and he asks oh and i know you've thought

32:24

about chapter value to reward counterfactual um impacts like making a good attempt that

32:31

falls short of someone else's i think having people engage in public discourse about chaplain value

32:37

attributions could really help reward people who almost made it and thereby incentivize more attempts to do good

32:44

could you comment on whether something like that seems good or what challenges it would raise

32:52

i think that the main challenge uh

32:57

that i'm worried about is this thing that i alluded to about uh trying to sell impact when

33:03

uh there's unclear ownership of the rights i think that it could create

33:08

adversarial uh incentives for people to uh overstate their contributions and

33:14

understate uh uh those of others however i i see i see like the prize

33:20

that you're pointing at and i i think maybe there could be something good there i

33:25

feel more like oh this is worth more thought not i know

33:32

how to make this work um i'm in the moment mildly pessimistic about

33:37

uh getting it to work in a straightforward way but there are lots of uh

33:44

possibly more complex implementations which might dodge uh

33:49

many or more all of the issues oh and do you think bottling and consuming is required to make impact

33:56

certificates work why can't consumption be a time bandwidth oriented i.e the current holder of the certificate like with art

34:03

markets i i think that this is uh

34:09

a pretty significant bifurcation in how you want to uh conceive up and set up impact markets i think that the

34:17

uh world where you say you treat it like the art markets and you say

34:23

the current holder is deriving value makes some sense but it has

34:28

it loses significant amounts of cleanness relative to the uh one where uh which has consumption and it i think

34:35

it may lose uh significant amounts of the benefit so some of the cleanness that it loses is

34:42

uh you can no longer just look at something and say oh how good was this as as a way of informing uh how

34:49

much you should care about uh buying it you sort of want to look at well how much of the value has already

34:54

depreciated um uh and uh

35:01

uh you're going to be baking in some expectations of well how much

35:09

uh society uh seems to care about this thing but it

35:15

reduces the ability to say well we want to to use this as a kind of like consequentialist

35:21

mechanism um i think that it

35:27

also probably makes it significantly harder to set up in a way that gives you

35:33

anything like tax deductibility um uh uh for buying these things i think that uh

35:40

my understandings of like the principles of how tax deductibility is supposed to work uh not anchored on uh the laws of

35:47

any any one country the idea is when you give money to a thing for which you do

35:52

not receive like material benefit that's the point at which you should be given the tax deduction and i think that

35:59

uh the fine art model of impact certificates never has such a moment um

36:06

and therefore it is hard to work out how the tax deduction should

36:11

work i think that the uh version with consumption does have such a moment and that's better

36:18

and then i also think that the

36:23

fine art model makes it harder to uh

36:29

incentivize profit motivated uh speculators and analysts to enter into

36:35

the market um because you ex uh in this model you expect by

36:42

default uh most certificates to depreciate with time uh maybe there'll be some which go up in value because uh

36:49

uh uh people just realized how great the thing was and so you can still get some of this but essentially there will be a

36:55

significant tax uh on anyone who is uh trying to enter into this uh from the

37:01

perspective of well uh i'm not gonna be the person ultimately like buying these and wanting to hold the impact but i can

37:07

add good information to the system and i i can earn a profit for doing that and so i'll i'll come and uh help things out

37:15

and and those people are just a bit more pushed out of the system which means that i expect to

37:20

get uh uh less of the benefits of relatively efficient markets uh out of that all of

37:27

that said you totally can do a version uh without consumption um

37:35

there's also the uh uh the issue of maybe you're more likely to just get bubbles um uh in in the version

37:41

without consumption uh because there's nothing uh uh propping up the uh uh uh the price of things

37:49

um but the yeah the original proposal for certificates of impact uh by paul cristiano and catcher grace uh didn't

37:55

have consumption uh i i feel

38:01

less optimistic about it doing really good things but maybe i'm wrong oh and do you know of anyone building

38:07

the market you described are you planning to and what support would be most helpful i am personally not planning to but i

38:15

know uh a dennis uh dresher who

38:21

uh uh helps me with some of the thinking about

38:27

the uh the voici standards has been uh thinking about what would be needed to

38:34

get a fairly minimal uh implementation of this uh set up on

38:40

top of uh nfts uh uh in some existing

38:46

chains um actually i haven't touched base with him recently enough to know what the current

38:51

status of that is i think maybe we were thinking oh we should publish the blog post on furici and we haven't quite got

38:57

that great well we would love to keep uh apprised of this research and there's some excitement in the chat about this

39:03

idea and i'd like to thank owen again for his time

39:14

you